HOMEWORK 7.3 · UNIT 7 · NC.8.SP.4 · NC.8.SP.3 · NC.8.SP.1

Two-Way Frequency Tables

Targets: SP.4-2, SP.4-3, SP.4-1, SP.3-2, SP.4-4, SP.1-5, SP.4-7, SP.4-8

Name _____ Date _____ Block _____

Remember: A **two-way table** counts data in two categories. Add across for row totals. Add down for column totals. **Joint** = a count inside the table. **Marginal** = a row or column total. **Relative frequency** = count ÷ total. Ex: 32 of 40 students like pizza: $32 \div 40 = 0.8 = 80\%$.

SCORE 3.0 Approaching · the basics

Targets: SP.4-2, SP.4-3, SP.4-1, SP.3-2

1 A school asked 180 students about breakfast. How many **7th graders skip** breakfast? SP.4-2

	Eats breakfast	Skips	Total
6th grade	42	18	60
7th grade	35	25	60
8th grade	28	32	60
Total	105	75	180

2 Use the breakfast table. How many students skip breakfast in all? Is it a joint or a marginal frequency? SP.4-2

3 Use the breakfast table. What percent of **all** students eat breakfast? Round to the nearest tenth. SP.4-3

Ⓐ 58.3% Ⓑ 41.7% Ⓒ 70% Ⓓ 105%

4 50 adults chose tea or coffee. 30 are women. 18 women chose coffee. 8 men chose tea. Find *a*, *b*, *c* and *d*. SP.4-1

	Tea	Coffee	Total
Women	a	18	30
Men	8	b	c
Total	d	30	50

5 A gym costs $y = 30x + 25$ dollars. x = months. What does 25 mean? REVIEW 7.2 SP.3-2

Ⓐ $25 to sign up Ⓑ $25 each month Ⓒ 25 months Ⓓ $30 to sign up

SCORE 4.0 Proficient · on target

Targets: SP.4-4, SP.1-5

SP.4-4

6 Use the breakfast table. What percent of **8th graders** skip breakfast? Round to the nearest tenth.

SP.4-4

7 **Dan** made a mistake. Circle it. Explain it. Fix it.

DAN'S WORK

What percent of students who skip breakfast are 6th graders?

18 ÷ 60 = 30%

REVIEW 7.1 SP.1-5

8 A scatter plot's dots go down from left to right, but they are spread out. Describe it.

Ⓐ Strong positive Ⓑ Weak positive Ⓒ Strong negative Ⓓ Weak negative

SCORE 5.0 Mastery · go deeper

Targets: SP.4-7, SP.4-8

SP.4-7

9 Use the breakfast table. Is there an association between grade and skipping breakfast? Use percents.

6th grade: _____ %. 8th grade: _____ %. So _____ .

SP.4-8

10 Suppose 40% of every grade skipped breakfast. What would that tell you about grade and breakfast?

It would mean _____ because _____ .